NATURAL LANGUAGE PROCESSING

Project Title

SafeSignal: Detecting Hidden Drug Side-Effects from Patient Reviews using NLP

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Team Members	Sanobar Latif Shaikh - 25030421039
Faculty	Dr. Nabanita Choudhury, Assistant Professor

SYMBIOSIS ARTIFICIAL INTELLIGENCE INSTITUTE (SAII)

Constituent of Symbiosis International (Deemed University)
Lavale, Pune - 412115

1. Abstract

Clinical trials are run on limited groups of people for limited periods, so many adverse drug reactions (ADRs) only show up once a medicine is in everyday use. Patients often describe these side-effects on health forums in their own casual words, but that information is rarely analysed. SafeSignal is an NLP system that reads such patient-written posts and automatically picks out Drug, ADR, Disease, Symptom and Finding mentions, acting as a simple early-warning tool for drug safety.

The project uses the CADEC (CSIRO Adverse Drug Event Corpus), which contains 1,250 annotated forum posts from AskaPatient.com about 12 drugs. A blank English spaCy pipeline with tok2vec and ner components was trained on 1,000 posts and tested on 250. On the test set the model detects Drug entities well (F1 = 0.91) and ADR entities reasonably (F1 = 0.65), but performs poorly on rarer classes such as Disease (0.18), Symptom (0.14) and Finding (0.35), mainly because of class imbalance. A formal-versus-informal audit also showed that Disease detection falls from 0.26 to 0.00 on informal text. The work shows that useful ADR signals can be mined from patient language, and also shows clearly where the model needs to improve before real-world use.

2. Introduction

Before a drug is approved it goes through clinical trials, but these involve a limited number of volunteers who are followed for a limited time. Rare, delayed or long-term side-effects often go unnoticed until thousands of people have used the drug. Regulators such as the FDA and WHO therefore run pharmacovigilance programmes that keep monitoring drug safety after launch.

At the same time, patients talk openly about their experiences online. A person might write that they started a medicine and soon after got muscle pain or trouble sleeping, and these posts are a large, free and constantly growing source of safety information. The difficulty is that this text is unstructured, informal and full of spelling mistakes, so it cannot be searched or counted easily by hand.

This is where NLP is useful. Named Entity Recognition (NER) can read raw text and label the drug names and the reactions linked to them, turning free text into structured data that can be counted and monitored. We chose this problem because it has real social value, it uses real-world messy language, and it lets us study both what NLP does well and where it breaks.

3. Problem Statement and Objectives

Problem statement: Adverse drug reactions reported informally by patients on online forums are valuable but remain largely unanalysed. The aim is to build an NLP pipeline that automatically extracts Drug, ADR, Disease, Symptom and Finding entities from patient-written text so that hidden ADR signals can be detected at scale.

Objectives:

- To parse and prepare the CADEC corpus (raw posts and BRAT annotations) for NER training.
- To train a spaCy NER model that recognises five entity types in patient text.
- To evaluate the model using entity-level precision, recall and F1-score, both overall and per class.
- To audit the model by comparing its behaviour on formal and informal writing.
- To identify the main failure cases and suggest realistic improvements.

4. Dataset Description

The project uses the CADEC (CSIRO Adverse Drug Event Corpus), which is publicly available through the CSIRO data portal (data.csiro.au). It is built from posts written by real patients on the AskaPatient.com forum.

Property	Details
Dataset	CADEC - CSIRO Adverse Drug Event Corpus
Source	AskaPatient.com forum posts, via data.csiro.au
Size	1,250 patient posts
Drugs covered	12 drugs (e.g., Lipitor, Voltaren, Arthrotec)
Annotation format	BRAT entity spans (.ann files linked to .txt posts)
Entity labels	Drug, ADR, Disease, Symptom, Finding
Language	English (informal, patient-written)
Train / test split	1,000 / 250 documents (80/20, random seed 42)

Table 1: Summary of the CADEC dataset used in this project.

Two characteristics of the data are important. First, the classes are imbalanced: Drug and ADR mentions are far more common than Disease, Symptom and Finding mentions, which later shows up clearly in the results. Second, the posts are written casually, with inconsistent punctuation, spelling variations and short fragments, which makes tokenisation and annotation alignment harder than with clean text.

5. Data Preprocessing

The raw data comes as plain text posts with separate annotation files, so it had to be converted before training. The steps were:

- Parsing: each raw .txt post was read together with its matching .ann annotation file to collect the entity labels and their character positions.
- Span conversion: character-offset entity spans were converted into spaCy-compatible token spans using char_span(), because spaCy trains on token boundaries rather than raw characters.
- Overlap filtering: overlapping entity spans were removed using spaCy's filter_spans(), since NER training cannot handle two labels covering the same words.

- Train/test split: the 1,250 documents were randomly divided into 1,000 for training and 250 for testing (80/20, seed 42) so results can be reproduced.

About 4% of the entity annotations had to be skipped because the annotated boundaries did not line up with spaCy's tokens. This is a known problem with informal forum text where punctuation is used inconsistently. We did not apply lowercasing, stop-word removal, stemming or lemmatisation, because NER relies on the original word forms and casing (for example, drug names are often capitalised) and removing words would break the character offsets of the annotations.

6. Feature Representation / Embeddings

The model does not use hand-built features such as Bag-of-Words or TF-IDF. Instead, it uses spaCy's built-in tok2vec layer, which learns vector representations of tokens as part of training. This layer looks at each word together with its surrounding words, so the representation of a token depends on its context. That is useful here because the same kind of word can be a drug in one sentence and part of a reaction in another.

We chose this approach because the pipeline is trained from scratch on CPU. Learned token vectors keep the model small and fast, avoid the need for large pretrained files, and are trained directly for the NER task. The trade-off is that a model trained from scratch on 1,000 documents has seen far less language than a pretrained transformer, which affects how well it copes with rare words and misspellings (see Section 11).

7. Proposed Methodology / System Architecture

SafeSignal follows a simple end-to-end flow. A patient post is taken as raw text, passed through the trained NER model, and returned with each drug and reaction highlighted as a labelled entity. The overall pipeline is shown in Figure 1.

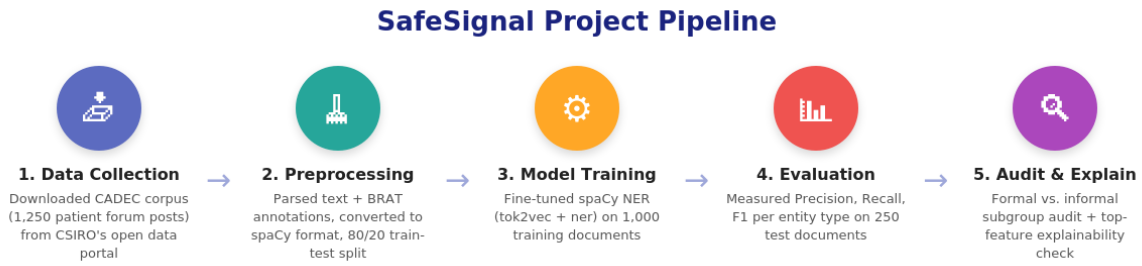


Figure 1: SafeSignal project pipeline from raw patient post to tagged entities.

The workflow has three stages: (1) Input - raw forum text from the CADEC posts; (2) Processing and model - tokenisation followed by the fine-tuned spaCy NER model; (3) Output - entities flagged as Drug, ADR, Disease, Symptom or Finding. No manual review is needed between stages.

8. NLP Model / Technique Used

The technique used is Named Entity Recognition with spaCy. The model starts from a blank English pipeline containing two components: tok2vec, which creates context-aware token vectors, and ner, which predicts the entity label of each token span. It was trained from scratch on the CADEC training documents using spaCy's command-line training pipeline, and the setup was kept light so that it could run on a CPU.

Setting	Value
Framework	spaCy (blank English pipeline)
Components	tok2vec + ner
Training data	1,000 documents
Test data	250 documents
Epochs	20
Training loss	12,254 to 3,137
Hardware	CPU only
Evaluation	Entity-level Precision, Recall, F1

Table 2: Model and training configuration.

Over 20 epochs the training loss dropped from 12,254 to 3,137 (Figure 2), which shows that the model was learning the annotation patterns steadily.

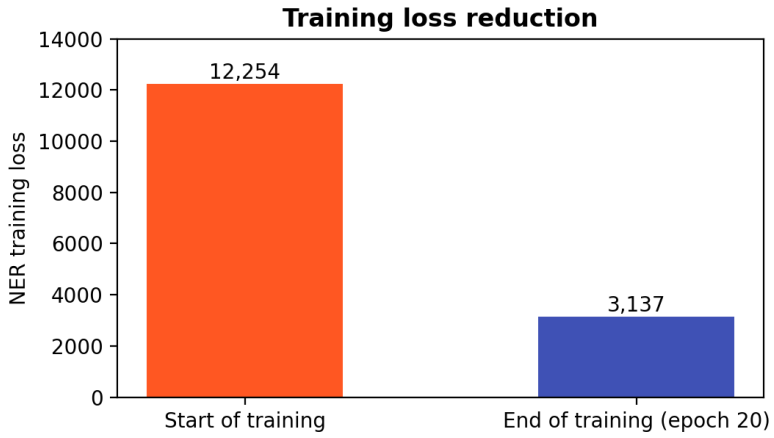


Figure 2: Reduction in NER training loss over 20 epochs.

9. Implementation

The system was implemented in Python using spaCy as the main NLP framework. The work is split into a few clear modules:

- Data loader: reads the .txt posts and .ann annotation files from CADEC.
- Converter: maps character offsets to spaCy token spans and filters overlaps.
- Trainer: runs spaCy's CLI training with the tok2vec + ner configuration.
- Evaluator: computes entity-level precision, recall and F1 for each class and overall, and splits results into formal and informal subsets for the audit.
- Predictor: loads the trained model and tags new, unseen sentences.

No separate web application was developed; the system is used through code and notebooks. As a check that the trained model works, it was tested on a new sentence it had never seen:

"I took Lipitor and started having severe muscle pain and memory problems."

Text span	Predicted label
Lipitor	Drug
severe muscle pain	ADR
memory problems	ADR

Table 3: Sample prediction on an unseen sentence.

The model separates the drug name from the reactions it may have caused, even though the sentence is casually worded.

10. Results and Evaluation

The model was evaluated on the 250 held-out test posts using entity-level precision, recall and F1-score. Table 4 gives the per-class results and Figure 3 shows the F1 scores graphically.

Entity	Precision	Recall	F1
Drug	0.94	0.88	0.91
ADR	0.65	0.65	0.65
Finding	0.43	0.29	0.35
Disease	0.33	0.12	0.18
Symptom	0.50	0.08	0.14

Table 4: Test set performance by entity type.

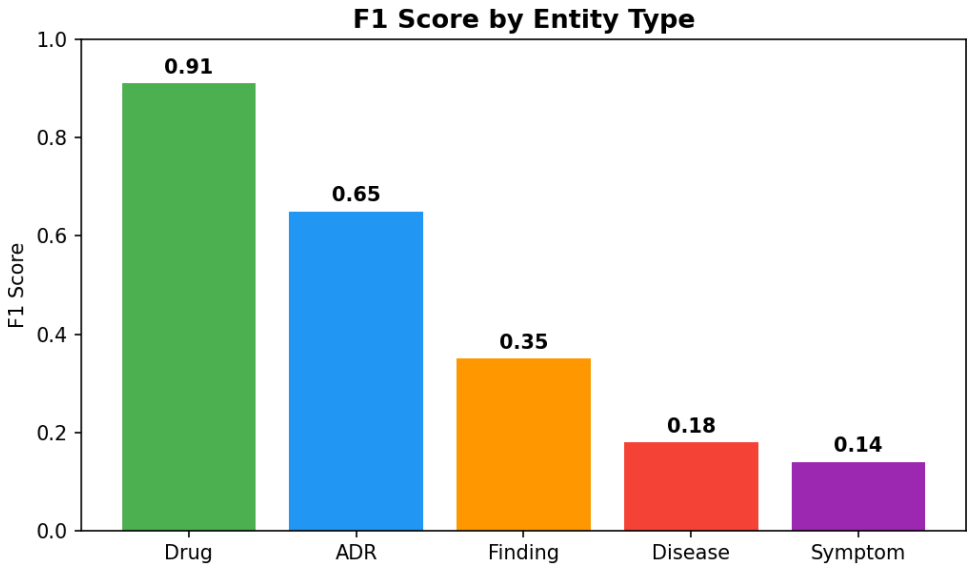


Figure 3: F1-score by entity type. Drug and ADR are detected far more reliably than the rarer classes.

Drug is the strongest class (F1 0.91), which makes sense because drug names are specific, repeated and often capitalised. ADR, the most important class for drug safety, reaches an F1 of 0.65 with balanced precision and recall, so it is useful but not yet reliable enough to work alone. Disease, Symptom and Finding perform much worse. Symptom has a precision of 0.50 but a recall of only 0.08, meaning that the few things the model labels as Symptom are often right, but it misses almost all of them. This pattern points to a lack of training examples for these labels.

Formal vs. informal audit. To check how the model behaves on different writing styles, the test posts were split into formal (clearly written) and informal (casual, misspelled) groups and scored separately (Figure 4).

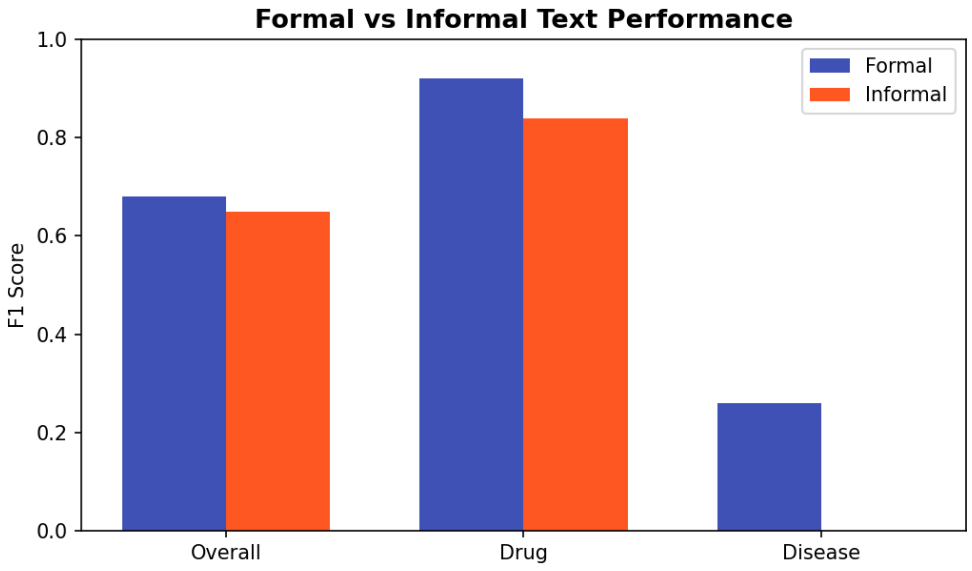


Figure 4: Formal vs. informal text performance (F1-score).

Entity	Formal F1	Informal F1
Drug	about 0.92	about 0.84
Disease	0.26	0.00
Overall	about 0.68	about 0.65

Table 5: Formal vs. informal F1 (values read from Figure 4; Disease values as reported in the audit).

Overall performance drops only slightly on informal text, and Drug detection falls by roughly 8%. Disease detection, however, collapses from 0.26 to 0.00. In other words, the model is weakest on exactly the kind of language that most patients actually use.

11. Error Analysis / NLP Failure Analysis

The metrics show where the model fails; the reasons below explain why. These come from the audit results and the nature of the data.

- Class imbalance: Disease, Symptom and Finding have far fewer training examples than Drug and ADR. Symptom recall of 0.08 shows the model barely learned this class.
- Informal language: on informal posts Disease F1 fell to 0.00. Casual wording, slang and spelling variations of medical terms look different from anything the model saw in training.
- Blurry label boundaries: an expression such as "muscle pain" could reasonably be an ADR, a Symptom or a Finding depending on context. Because these categories overlap, the model often picks the more common label (ADR), which hurts the rarer ones.
- Tokenisation misalignment: about 4% of annotations were lost because informal punctuation broke the token boundaries, so the model never saw some of the examples at all.
- Small, from-scratch model: with only 1,000 training posts and no pretrained knowledge, the model cannot recognise unseen medical words or misspellings.

A typical failure of this kind would be a post saying a patient has "bad stomache pains" or a condition written in a casual, misspelled form: the unusual spelling makes the word unfamiliar, and the model either skips it or gives it the wrong label.

12. Ethical Considerations and Limitations

- Medical safety: the output is not a diagnosis. A missed or wrongly tagged reaction could mislead anyone who relied on it, so the system should only support human reviewers, never replace them.
- Privacy: forum posts are personal health information. CADEC is a public research dataset, but any real deployment must handle patient text carefully and avoid identifying individuals.
- Bias and representation: the data comes from one forum and 12 drugs, so it reflects the people who use that site and may not represent other ages, regions or languages. The model also performs worse on informal writing, which could disadvantage patients who write casually or are less fluent in English.

- Dataset limitations: small size, class imbalance, English only, and about 4% of annotations dropped during preprocessing.
- Model limitations: trained from scratch on CPU, with weak results on Disease, Symptom and Finding, and no link yet between a reaction and the specific drug that caused it.

13. Conclusion

In this project we built SafeSignal, an NER-based system that reads patient forum posts and extracts drugs and reported reactions. Using spaCy and the CADEC corpus, the model reached an F1 of 0.91 for Drug and 0.65 for ADR, which are the two most important categories for drug-safety monitoring. The project therefore met its main objectives of preparing the data, training the model, evaluating it per class and auditing it on formal and informal text.

The audit also gave an honest picture of the limits: rare classes are weak and Disease detection fails completely on informal text. The main lesson is that a good score on a clean benchmark does not guarantee good behaviour on real patient language, and that error analysis is as important as the headline metric.

14. Future Scope

- Train on a larger and more varied dataset to fix the class imbalance for Disease, Symptom and Finding.
- Use transformer models such as BioBERT, which already know medical vocabulary and handle misspellings better.
- Add text normalisation or spelling correction for informal patient language before tagging.
- Link each ADR to the drug that caused it using relation extraction.
- Build a simple web interface so that non-technical users can paste a post and see highlighted entities.
- Extend to other forums and languages, including code-mixed text.

15. References

[1] Karimi, S., Metke-Jimenez, A., Kemp, M., and Wang, C. (2015). CADEC: A corpus of adverse drug event annotations. Journal of Biomedical Informatics, 55, 73-81.

[2] CSIRO. CADEC - CSIRO Adverse Drug Event Corpus. CSIRO Data Access Portal, https://data.csiro.au

[3] Honnibal, M., and Montani, I. spaCy: Industrial-strength Natural Language Processing in Python. https://spacy.io

[4] spaCy Documentation: Training pipelines and models. https://spacy.io/usage/training

[5] Lee, J., et al. (2020). BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics, 36(4), 1234-1240.

16. GitHub Repository

GitHub Repository Link- github.com/sanobarshaikhbsc2025-cyber/SafeSignal

17. Team Contribution

Team Member	PRN	Specific Contribution
Sanobar Latif Shaikh	25030421039	Complete project work: dataset study and preprocessing, model training, evaluation, formal vs. informal audit, result visualisations, pitch deck and this report.

18. Annexure

Annexure A Additional example. Input: "Took Lipitor for 2 years, now I have muscle pain and can't sleep most nights." Expected output: Lipitor tagged as Drug and the reactions tagged as ADR.

Annexure B: Project write-up. The build component write-up (SafeSignal_Writeup.docx) contains the original problem statement, dataset notes and result summary.